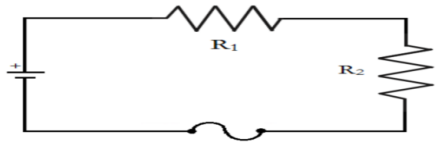
UNIT-I

DC & AC CIRCUITS

Electric circuit

An Electric circuit is an interconnection of various elements in which there is at least one closed path in which current can flow. An Electric circuit is used as a component for any engineering system. The performance of any electrical device or machine is always studied by drawing its electrical equivalent circuit.



Electrical Parameter	Measuring Unit	Symbol
Voltage	Volt	V or E
Resistance	Ohm	R or Ω
Capacitance	Farad	C
Charge	Coulomb	Q
Inductance	Henry	L or H
Power	Watts	W
Impedance	Ohm	Z
Frequency	Hertz	Hz
Conductance	Siemen	G or ℧

Ohm's law

Ohm's law states that , at constant temperature the voltage across a conductor is directly proportional to the current flowing through it.

$$V = I \cdot R$$

$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$

Limitations of ohm's law:

1. Ohm's law does not apply to unilateral electrical components such as diodes as well as transistors even though they only permit current just to flow in one way.
2. Voltage level will not be constant with respect to time for non-linear electrical components with properties such as capacitance, resistance, and so on, making Ohm's law problematic to apply.

Voltage

The voltage is the difference in potential between two points in an electric field. To determines the voltage between any two points, both a static electric field and a dynamic electromagnetic field is considered. The mathematical representation of voltage is as follows:

- V = IR
- V = Voltage in volts
- I = Current in amperes
- R = Resistance in ohms

Current

The flow of electric charge is referred to as **"electric current."** Current is a measure of the movement of charged particles per unit of time. Although, once again, these particles are typically electrons. According to Ohm's law, the current is the ratio of the potential difference and the resistance. Thus, the current formula is given by: I = V/R

I - represent current in Ampere,
V - is the potential difference in Volt
R - is the resistance in Ohm(Ω).

Power

Electric power is the rate at which electrical energy is transferred by an electric circuit (or) The product of voltage and current in electrical circuits.

The electric power is given by **P = VI**, where V is the potential difference, I is the electric current and P is the electric power.

Energy

Energy is defined as the capacity to do work. Formula. The energy stored in an object due to its position and height is known as potential energy and is given by the formula: **P.E. = mgh**.

m denotes the object's mass, g denotes gravitational acceleration, and h denotes height.

The formula to determine the Kinetic energy of an object is – **K.E = ½ mv²**

[where,m= Mass of the object, V = The velocity of the object]

Active and passive components

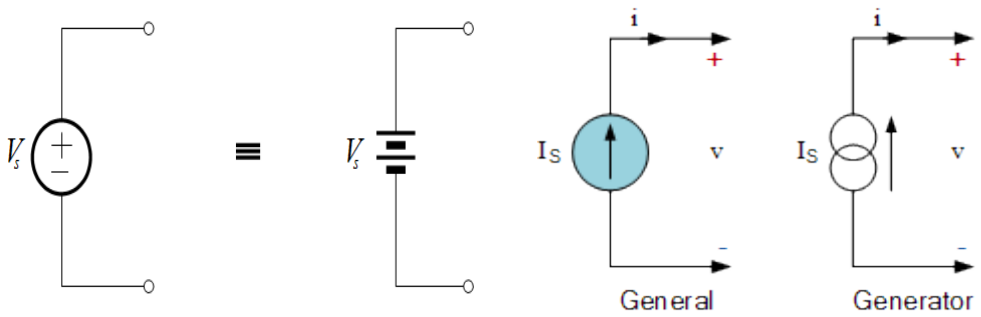
Active and passive components form the two main types of electronic circuit elements. An active component supplies energy to an electric circuit, and hence has the ability to electrically control the flow of charge. A passive component can only receive energy, which it can either dissipate or absorb.

Active components

An **active component** is an electronic component which supplies energy to a circuit. Active elements have the ability to electrically control electron flow (i.e. the flow of charge). All electronic circuits must contain at least one active component.

Common examples of active components include:

1.Voltage sources
2.Current sources (e.g. DC current source)
3.Generators (such as alternators and DC generators)
4.All different types of transistors (such as bipolar junction transistors, MOSFETS, FETs, and JFET)
5.Diodes (such as Zener diodes, photodiodes, Schottky diodes, and LEDs)



(1) Voltage Sources

A voltage source is an example of an active component in a circuit. When current leaves from the positive terminal of the voltage source, energy is being supplied to the circuit. As per the definition of an active element, a battery can also be considered as an active element, as it continuously delivers energy to the circuit during discharging.

(2)Current Sources

A current source is also considered an active component. The current supplied to the circuit by an ideal current source is independent of circuit voltage. As a current source is controlling the flow of charge in a circuit, it is classified as an active element.

Passive Components

A **passive component** is an electronic component which can only receive energy, which it can dissipate, absorb or store it in an electric field or a magnetic field. Passive elements do not need any form of electrical power to operate. As the name 'passive' suggests – passive devices do not provide gain or amplification. Passive components cannot amplify, oscillate, or generate an electrical signal.

Common examples of passive components include:

- Resistors
- Inductors
- Capacitors
- Transformers

Resistors

A resistor is taken as a passive element since it cannot deliver any energy to a circuit. Instead resistors can only recieve energy which they can dissipate as heat as long as current flows through it. units of resistance is ohms

Resistor

Inductors

An inductor is also considered as passive element of circuit, because it can store energy in it as a magnetic field, and can deliver that energy to the circuit, but not in continuous basis. The energy absorbing and delivering capacity of an inductor is limited and transient in nature. That is why an inductor is taken as a passive element of a circuit. Units of inductance is Henry (H)

Inductor

Capacitors

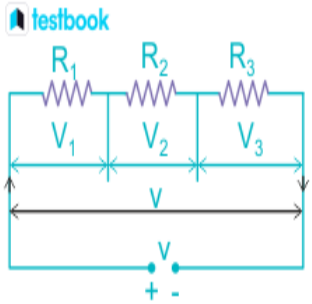
A capacitor is considered as a passive element because it can store energy in it as electric field. The energy dealing capacity of a capacitor is limited and transient – it is not actually supplying energy, it is storing it for later use. Capacitor units is Farad (F)

Capacitor

Resistances (Or Resistors) In Series

When two or more resistances are joined end to end so that same current flows through each of them in turn, they are said to be connected in series. Here, the total resistance is equal to the sum of the individual resistances.

$$R = R_1 + R_2 + R_3$$

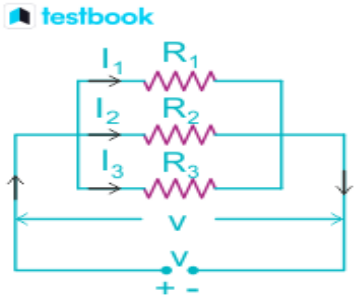


DC Series Circuit

Resistances (Or Resistors) In Parallel

When two or more resistances are connected across two points so that each of them provides a separate path for current, they are said to be connected in parallel. Here the reciprocal of their combined resistance is equal to the sum of the reciprocals of the individual resistances.

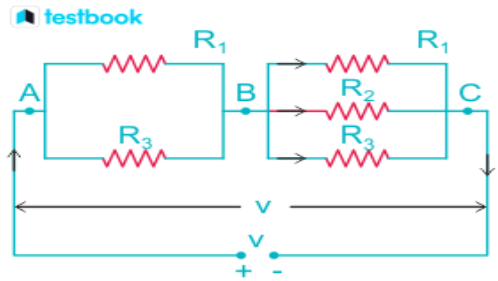
The total resistance becomes $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$



DC Parallel circuit

Series-Parallel Circuit

It is a circuit in which the series and parallel circuits are connected in series.



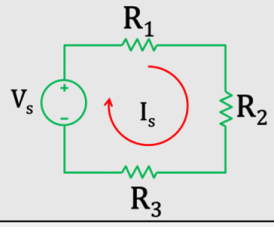
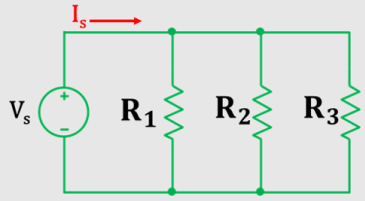
DC Series Parallel Circuit

Total resistance in the entire circuit can be determined by first finding out the sum of the equivalent resistance of A and B.

$$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1+R_2}{R_1R_2}$$

$$R_{BC} = \frac{R_3R_4R_5}{R_3R_4+R_4R_5+R_5R_3}$$

$$R = R_{AB} + R_{BC}$$

	Series	Parallel
How it looks		
Voltage	$V_s = V_1 + V_2 + V_3$ $V_1 = I_sR_1; V_2 = I_sR_2; V_3 = I_sR_3$	$V_s = V_1 = V_2 = V_3 = I_sR_{eq}$
Current	$I_s = I_1 = I_2 = I_3 = \frac{V_s}{R_{eq}}$	$I_s = I_1 + I_2 + I_3$ $I_1 = \frac{V_s}{R_1}; I_2 = \frac{V_s}{R_2}; I_3 = \frac{V_s}{R_3}$
Resistance	$R_{eq} = R_1 + R_2 + R_3$	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
Features	If one components burns, the circuit becomes inactive and current flow stops	If one component burns current stops only through that branch rest part works fine

Kirchhoff's Laws

There are some simple relationships between currents and voltages of different branches of an electrical circuit. These relationships are determined by some basic laws that are known as **Kirchhoff laws** or more specifically **Kirchhoff Current and Voltage laws**. These laws are very helpful in determining the equivalent electrical resistance or impedance (in case of AC) of a complex network and the currents flowing in the various branches of the network. These laws are first derived by **Guatov Robert Kirchhoff** and hence these laws are also referred as Kirchhoff Law.

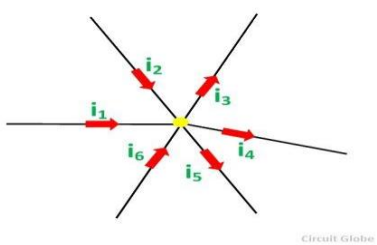
Kirchhoff's Current Law

According to the **Kirchhoff's Current Law**, the algebraic sum of the currents entering a node must be equal to the algebraic sum of the currents leaving the node in an electrical network. (Or)

It states that" the algebraic sum of all the currents at any node point or a junction of a circuit is zero"

$$\Sigma I = 0$$

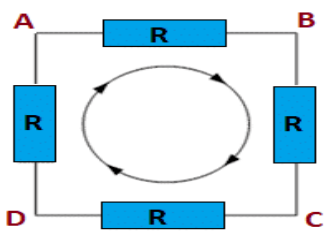
$$\mathbf{i_1 + i_2 - i_3 - i_4 - i_5 + i_6 = 0}$$



Kirchhoff's Voltage Law

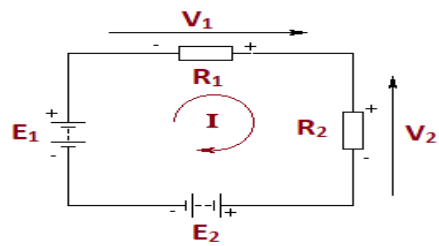
Kirchhoff's Voltage Law states that the algebraic sum of the voltages (or voltage drops) in any closed path of a network that is transverse in a single direction is zero. In other words, in a closed circuit, the algebraic sum of all the EMFs and the algebraic sum of all the voltage drops (product of current (I) and resistance (R)) is zero.

$$\mathbf{\Sigma V = 0}$$



$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$

(a)



$E_1 + V_1 - V_2 - E_2 = 0$

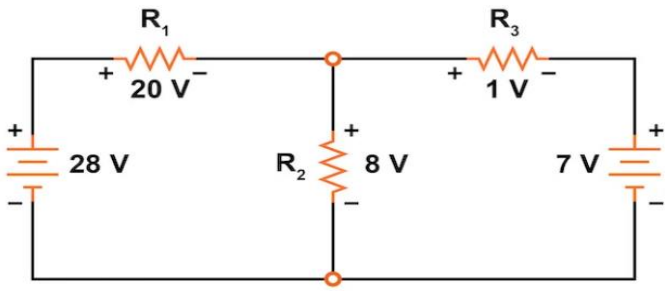
$E_1 + V_1 = V_2 + E_2$

(b)

Superposition Theorem:

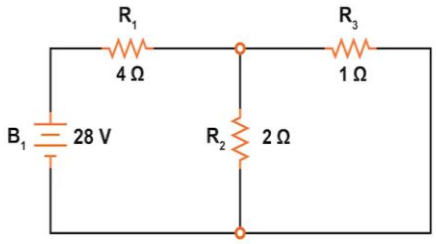
In any linear, bilateral, multi source network the response in any element is equal to the algebraic sum of the responses obtained by each source acting alone while all other sources are set equal to zero.

$$I = I_1 + I_2 + - - - - + I_n$$

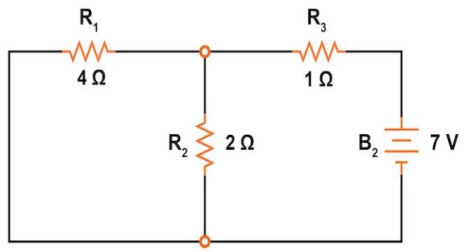


Procedure:

1. Find the I_1 in a particular branch by considering source 1 is activating and Deactivate source 2(s/c).



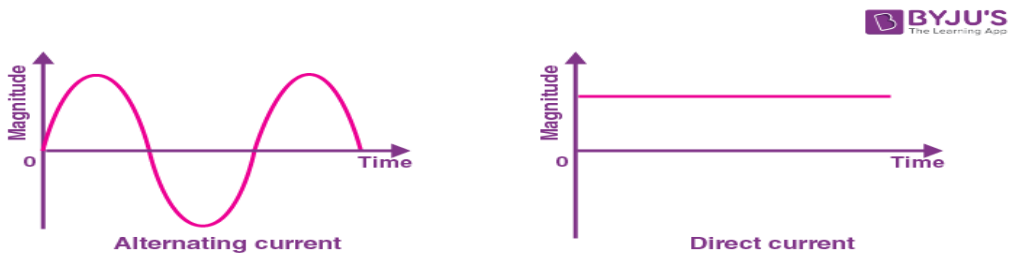
2. Find the I_2 in a particular branch by considering source 2 is activating and Deactivate source 1(s/c).



3. Find the total current $\mathbf{I = I_1 + I_2}$

Alternating current (AC)

An alternating current can be defined as a current that changes its magnitude and polarity at regular intervals of time. It can also be defined as an electrical current that repeatedly changes or reverses its direction opposite to that of Direct Current or DC which always flows in a single direction as shown below.



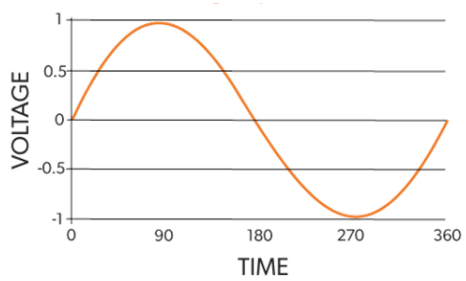
Phase

In electronic signaling, a phase is **the position of a wave at a point in time (instant) on a waveform cycle**. It provides a measurement of exactly where the wave is positioned within its cycle, using either degrees (0-360) or radians (0-2π). One radian of a phase is equal to about 57.3 degrees.

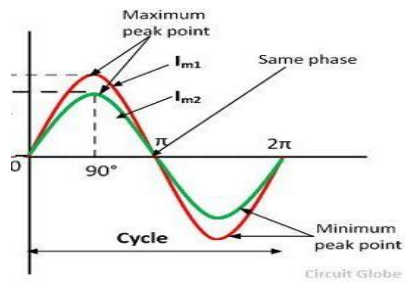
Phase difference

Phase difference is the difference in phase angle between two sinusoids or phasors. The phase difference is the difference in the phase angle of the two waves. Path difference is the

difference in the path traversed by the two waves. The relation between phase difference and path difference is direct. They are directly proportional to each other.



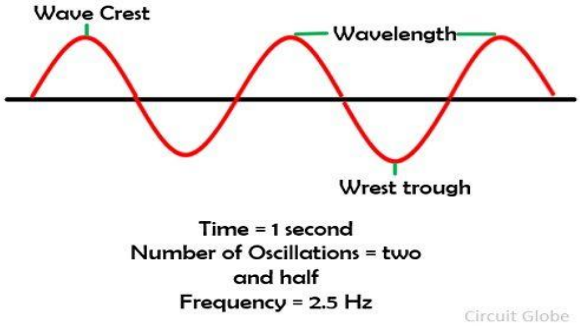
(i) Phase



(ii) Phase difference

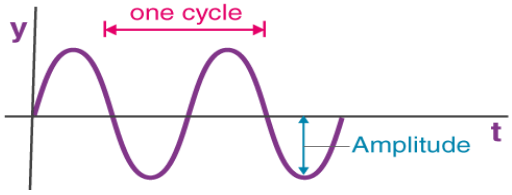
Frequency

Frequency is **the rate at which current changes direction per second**. It is measured in hertz (Hz), an international unit of measure where 1 hertz is equal to 1 cycle per second. Hertz (Hz) = One hertz is equal to one cycle per second. Cycle = One complete wave of alternating current or voltage.



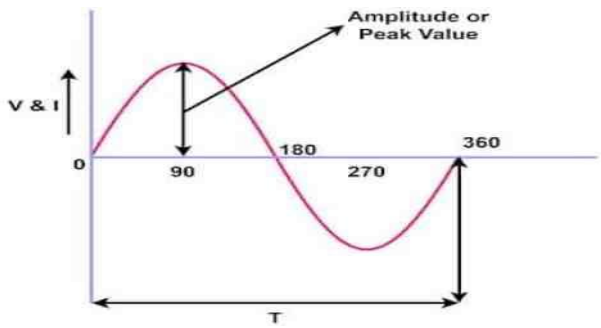
Time Period

The time period is **the time taken by a complete cycle of the wave to pass a point**. Angular frequency is angular displacement of any element of the wave per unit of time.



Peak value

Peak value is defined as the maximum value reached by an alternating quantity in one cycle is known as Peak value.



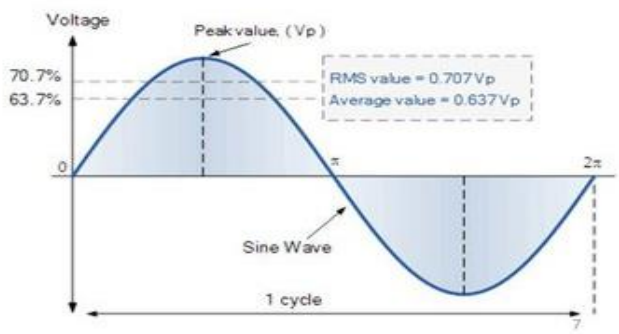
Average Value

- In a pure sine wave if the average value is calculated over the full cycle, the average value would be equal to zero as the positive and negative halves will cancel each other out.
- Then the average value is obtained by adding the instantaneous values of voltage over one half cycle only.

$$V_{av} = \frac{1}{\pi}\int_0^{\pi} V_p \sin\theta \; d\theta$$

$$V_{av} = \frac{V_p}{\pi}[-\cos\theta]_0^{\pi}$$

$$V_{av} = \frac{2V_p}{\pi} = 0.637V_p$$



RMS Value

For alternating electric current, RMS is equal to the value of the constant direct current that would produce the same power dissipation in a resistive load.

$$\text{rms value} = \frac{V_m}{\sqrt{2}}$$

The **RMS value** is the **effective value** of a varying voltage or current. It is the equivalent steady DC (constant) value which gives the same effect.

***effective value* or *DC-equivalent* value**

The **rms value** of a periodic function is defined as the square root of the mean value of the squared function.

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}$$

Form Factor

The form factor of an alternating current waveform (signal) is the ratio of the RMS (root mean square) value to the average value.

$$\text{Form Factor} = V_{rms}/V_{av}$$

$$\text{Form Factor} = \frac{I_{rms}}{I_{av}}$$

Peak Factor

Peak Factor is also known as Crest Factor or Amplitude Factor. It is the ratio between maximum value and RMS value of an alternating wave.

$$\text{Peak Factor} = \frac{I_m}{I_{r.m.s}} \text{ or } \frac{E_m}{E_{r.m.s}}$$

Waveform	RMS value	Crest factor	Form Factor
Sine wave	$\frac{1}{\sqrt{2}}$or 0.7071	$\sqrt{2}$or 1.414	1.11
Triangular Wave	$\frac{1}{\sqrt{3}}$ or 0.577	$\sqrt{3}$or 1.732	1.154
Square Wave	1	1	1
Saw tooth Wave	$\frac{1}{\sqrt{3}}$ or 0.577	$\sqrt{3}$ or 1.732	1.154

Real power (True Power)

Real power is expressed in watts and as so represents the actual energy converted from electrical energy to useful work. The calculation for real power is the product of the apparent power and the cosine of the angle between the voltage and current waveforms.

$$P=VI\ \cos\phi \text{ watts}$$

Reactive power

Reactive power is the part of complex power that corresponds to storage and retrieval of energy rather than consumption. On an AC power system, there are two kinds of power - real power that actually does work, and reactive power that enables transformers to transform, generators to generate, and motors to rotate.

$$Q=VI\ \sin\phi\ \text{VAr}$$

Apparent power

The total power flowing is known as the "apparent power" and is measured as the product of the voltage and current (V * I).

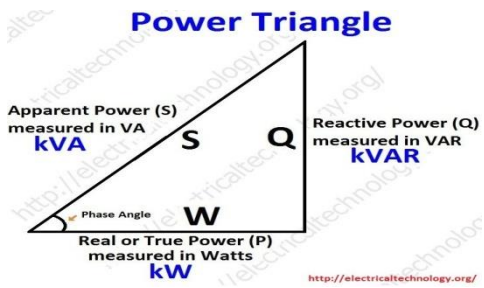
$$S= VI\ \text{va}$$

Power Factor

Power factor is defined as the cosine angle between voltage and current in ac circuits is known as power factor. (or) The ratio between Active Power and Apparent Power in volts-amperes is called power factor**.**

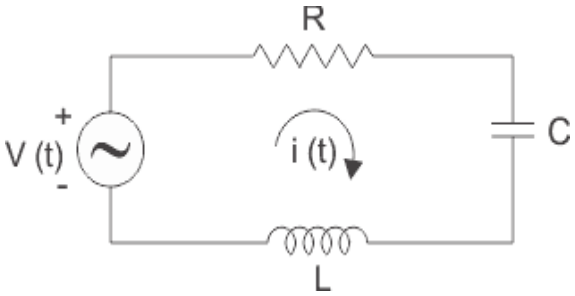
Power factor = **Cos ϕ**

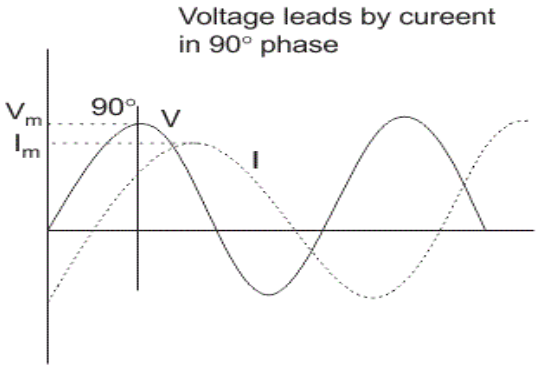
$$\text{Power factor} = \frac{\text{True power}}{\text{Apparent power}}$$

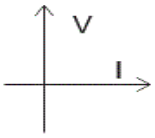


Phasor Diagram of Series RLC Circuit

The phasor diagram of series RLC circuit is drawn by combining the phasor diagram of resistor, inductor and capacitor. Before doing so, one should understand the relationship between voltage and current in case of resistor, capacitor and inductor.







1. **Resistor**
   In case of resistor, the voltage and the current are in same phase or we can say that the phase angle difference between voltage and current is zero.
2. **Inductor**
   In inductor, the voltage and the current are not in phase. The voltage leads that of current by 90° or in the other words, voltage attains its maximum and zero value 90° before the current attains it.
3. **Capacitor**
   In case of capacitor, the current leads the voltage by 90° or in the other words, voltage attains its maximum and zero value 0° after the current attains it i.e the phasor diagram of capacitor is exactly opposite of inductor.